BRIDGE_C_35 order7

order7 REPORT (descriptive notes; verdicts as in RESULTS.md)

PREREG sha256 251f3058…0e28 (unchanged). Script hashes: scripts/SHA256SUMS. Deviations are listed in NOTES.md, and all verdicts below are as recorded.

§0 Parity lemma

Check. $M^2 - 35P = 0$, verified numerically to 8.9e−30 (mpmath, dps 30; float64 gives 1.8e−14). Here M is the fixed 35-component law vector S-matrix and P is the involution $(\alpha,\beta) \mapsto (-\alpha,-\beta)$ with its phases.

Status. This is Category A by Weil-representation theory: $S^2 = Z$ acts as $\alpha \mapsto -\alpha$ with $e(-\mathrm{sig}/4) = 1$ for signature (1,1). The numerical check confirms it. The exact cyclotomic check promised in the PREREG was not done (NOTES 1).

Consequence (Category A, given $M^2 = 35P$). No nonzero P-even vector can obey the fixed weight-1 S-law $F(-1/\tau) = (\tau/\sqrt{35})\cdot M\cdot F(\tau)$. Applying the law twice gives $F = -PF$. The even double orthant is P-even, so **no completion of it can obey the fixed weight-1 law**. This explains the order6 failure. P-even objects are allowed at weight 0 and weight 2.

Candidate 1: non-Vignéras weight-1 object (Category B)

The candidate is $p_\lambda = \tfrac{1}{2}(1 + E1E2) + \lambda\cdot\exp(-\pi(\ell1^2 + \ell2^2))$, using the two orthant error-function terms from order6, $E1 = E(\sqrt{10}\cdot v1\cdot\sqrt{y})$ and $E2 = E(\sqrt{35}\cdot v2\cdot\sqrt{y})$, with $\ell1 = \sqrt{10}\cdot v1\cdot\sqrt{y}$ and $\ell2 = \sqrt{35}\cdot v2\cdot\sqrt{y}$. The best fit is $\lambda^* = -0.657 - 0.002i$.

quantity	value
residual at $\lambda = 0$	0.903
residual at λ^*	0.482
finite-difference check of $\partial_{\bar\tau}$ (product rule)	3.4e−9

Verdict: fails, as predicted by the lemma. The finite-difference shadow check passes, but it is only a calculus check: the $\partial_{\bar\tau}$ is not a sum of unary-theta products.

Candidate 2: weight-0 Siegel / Hecke–Maass objects for Q(√35)

The objects are $\Phi_k = \int e^{2\pi i k\mu/\mu_{tot}}\cdot\Theta_z\, d\mu$ over the unit-group arc from c1 to γc1. Here γ = [[−1,−7],[2,13]] (the unit 6+√35) and $\mu_{tot} = 2.478 \approx \log(6+\sqrt{35}) = 2.478$, consistent.

Law $\Phi(-1/\tau) = \varepsilon\cdot(1/\sqrt{35})\cdot M\cdot\Phi(\tau)$, weight 0, on 120 interior τ:

object	ε = +1	ε = −1	parity PΦ = Φ
Φ_0	5.5e−15	2.0	6.1e−15
Φ_1	1.2e−14	0.85	2.7e−14

Verdict: pass, with ε = +1, Category B. It also follows from Siegel's theorem for each Θ_z (Category A by citation). These are genuine P-even, weight-0 objects that obey the fixed 35-component law vector S-law.

Coupling to the even piece is inconclusive. The test asked whether Ê lies in the span of $y^{\mp 1/2}\Phi_k$. The residual is 4.3e−10, between the pass and fail lines. The interior-span test turned out to be non-discriminating (see Candidate 3), so no coupling claim is made.

Candidate 3: enlarged law space W_420 = span[Ê, Ĥ, Ĝ_O, Θ_E]

This space is separate from the 35-component law vector. Ê is the product-completed even piece, Ĥ the completed cone part, Ĝ_O the completed odd orthant pair, and Θ_E the holomorphic even series. The test used 660 interior τ.

- **First run (c3.py): void.** The positive control failed (residual 0.93). The cause was identically-zero companion columns, which were divided by their own ~1e−17 norms (NOTES 2).
- **Disclosed rerun (c3b.py), with near-zero columns dropped:**

target	residual, w = 0	residual, w = 1
Ĥ (positive control, w = 1)	2.0e−7	9.9e−9
Ê	1.6e−9	2.8e−10
Θ_E	4.3e−6	5.0e−7
Ŝ_ν (full, completed)	4.3e−8	4.1e−9

The numerical rank of W is 97, with singular values decaying smoothly through the cut (1.1e−12 → 7.1e−13). There is no gap. - **Verdict: inconclusive / void.** The positive control does not reach the 1e−10 pass line. Every target, including the wrong-weight Ĥ at w = 0, lands between 1e−10 and 1e−4. Holomorphic functions on a compact τ-patch are numerically low-rank in float64, so this design cannot separate "law holds" from "law fails". This is a design failure, not evidence either way.

Verdicts and the shadow

item	verdict
Parity lemma	holds (numerically to 1e−30; Category A by Weil-rep theory)
C1 (weight-1 non-Vignéras)	**fails**; it is ruled out in principle at weight 1 by the lemma
C2 (weight-0 Siegel/Hecke–Maass)	**objects pass** the fixed weight-0 law (ε = +1); coupling to the even piece **inconclusive**
C3 (W_420)	**void/inconclusive** (non-discriminating design)
Full-series shadow	**NOT located**

Appendix: order7 notes (disclosures; verdicts stand)

1. **Parity-lemma check.** PREREG §0 promised an exact check in Z[e(1/9800)]. What was done is an mpmath check at dps 30 (residual 8.9e−30) plus a float check. The lemma's Category A status rests on Weil-representation theory ($S^2 = Z$), not on an exact certificate.
2. **c3.py, first run: void.** The positive control failed because companion columns that vanish identically (for example $\hat{H}_j = 0$ by symmetry) were normalised by norms of about 1e−17. c3b.py (a new file, hashed before it ran) drops columns with norm below 1e−10·max. The void verdict of the first run stands.
3. **c3b: also non-discriminating.** The rerun's positive control reaches only 9.9e−9, against a pass line of 1e−10, and the singular values decay smoothly with no gap. The candidate-3 design is therefore inconclusive by construction. The candidate-2 coupling test shares the same flaw.
4. **Reading (descriptive).**
   - The even piece cannot be completed at weight 1 under the fixed law (lemma).
   - Weight-0 P-even modular objects of Hecke–Maass type, built from Siegel thetas over the arc of the unit $6+\sqrt{35}$, do exist and obey the fixed weight-0 law with ε = +1.
   - The open question is how a weight-1 even holomorphic series couples to them. Candidates are a weight-raising or lowering operator, a non-modular "mixed" object, or a T-orbit or level change that alters P.
5. **Process.** nice, OMP_NUM_THREADS=1, ulimit -v 3000000; peak memory per process well under 1 GB. No push, no external sends. New files only.